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Research advances in integrated traditional Chinese and Western medicine for the prevention and treatment of perioperative shivering

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Perioperative shivering (POS) is a common complication during anesthesia recovery and has traditionally been regarded as a simple thermoregulatory response. Modern research has revealed that it involves complex mechanisms, including activation of the neuro-immune-inflammatory axis and dysfunction of mitochondrial energy metabolism. Correspondingly, traditional Chinese medicine (TCM) attributes POS to Yang Qi deficiency, a concept that aligns theoretically with modern medical understanding of impaired thermogenesis mechanisms. For the prevention and treatment of POS, Western medicine primarily employs multi-target drugs, such as dexmedetomidine, and active temperature management strategies, while TCM emphasizes holistic regulation through methods like Shenfu injection and transcutaneous electrical acupoint stimulation. By adopting an integrated treatment model combining traditional Chinese and Western medicine, we have established a closed-loop management pathway encompassing “assessment–early warning–intervention” supported by AI-assisted evaluation and intelligent TCM diagnostic methods. This approach achieves an organic unity of modern precision medicine and the holistic concept of TCM.

KEYWORDS

body temperature regulation, dexmedetomidine, medicine, Chinese traditional, mitochondria, shivering

1 Introduction

Perioperative shivering (POS) is a common complication following anesthesia, with an incidence ranging from 60% to 80% during the recovery period (1). This condition not only causes significant patient discomfort but also triggers a range of physiological risks, including markedly increased oxygen consumption, hypoxemia, elevated risk of myocardial ischemia, potential exacerbation of postoperative pain, and delayed recovery (2). Therefore, effective prevention and management of POS have become critical components of the enhanced recovery after surgery (ERAS) framework aimed at improving surgical outcomes.

Perioperative shivering was traditionally viewed as a physiological response triggered by an elevated hypothalamic thermoregulatory set point to maintain core body temperature.

However, growing evidence indicates that its underlying mechanism is far more complex, involving a networked process across multiple systems. On the one hand, surgical trauma and anesthetic agents can activate the “neuro-immune-inflammatory axis,” driving an upward shift of the central thermoregulatory set point through signaling pathways such as prostaglandin E2 and the HMGB1/TLR4/NF- κ B cascade (3). On the other hand, peripheral skeletal muscle contraction during shivering relies heavily on mitochondrial energy metabolism. The functional integrity of electron transport chain complex I is crucial to meet the sharply increased ATP demand, and dysfunction of this complex is considered a core mechanism underlying the phenomenon of “shivering without effective heat production” (4).

From a theoretical perspective, these advances in modern medicine show a striking convergence with traditional Chinese medicine (TCM), which identifies “Yang Qi deficiency and vital gate fire decline” as the fundamental pathogenesis of shivering. Although Western medicine has developed targeted drugs like dexmedetomidine—which combines central sedative effects with multi-target anti-inflammatory properties—along with active temperature management protocols, the limitations of unimodal interventions remain evident (5, 6). In parallel, TCM—utilizing formulas and techniques such as Shenfu injection and transcutaneous electrical acupoint stimulation (TEAS) that embody the principle of “warming Yang and dispelling cold”—demonstrates unique advantages by regulating energy metabolism and neuro-immune function through holistic, multitarget regulation mechanisms.

In this context, it has become imperative to establish a comprehensive prevention and management framework that incorporates multidisciplinary integration and combines traditional Chinese and Western medicine. Over the past 5 years, advances in AI-assisted diagnostic technologies—such as computer vision-based myotremor analysis and intelligent tongue-pulse diagnostic systems—have substantially improved the early identification and monitoring of shivering. Meanwhile, pharmacological interventions have become increasingly targeted. For instance, the multi-pathway anti-shivering mechanisms of dexmedetomidine have been further elucidated, while non-drug therapies such as TEAS have demonstrated distinct advantages in regulating neuro-immune pathways and suppressing inflammatory responses. Furthermore, the integration of novel warming technologies, such as nanoparticle-based phase-change thermal garments, with AI-driven dynamic temperature management systems provides a more robust physical safeguard for intraoperative temperature management.

Therefore, this article aims to systematically review recent research advances in POS, providing a comprehensive analysis that spans underlying mechanisms to integrated clinical strategies. It explores the pathological interconnections between TCM and Western medicine regarding shivering pathogenesis and summarizes a multidimensional prevention and management system integrating “assessment–early warning–intervention.” The goal is to offer a novel theoretical framework and practical pathway to reduce the incidence of POS and improve patient outcomes.

2 Understanding the mechanisms of POS from traditional Chinese and Western medicine perspectives

2.1 Concept and assessment of shivering

Shivering is characterized by rapid, rhythmic muscle contractions in response to cold exposure or pathological conditions and represents a crucial mechanism by which the thermoregulatory system maintains core body temperature. POS commonly occurs during emergence from anesthesia, with a reported incidence of 60%–80%. Its development results not only from hypothermia but is also closely associated with the effects of anesthetic agents, surgical trauma, and stress responses (7). Shivering significantly increases oxygen consumption by 20%–35% and elevates the basal metabolic rate by 100%–300%. These physiological changes may lead to hypoxemia, acidosis, and even myocardial ischemia, thereby posing heightened risks, particularly for elderly patients and those with compromised cardiopulmonary reserve (8). Furthermore, shivering can exacerbate incisional pain, impede postoperative recovery, increase the risk of infection, and thereby adversely affect postoperative outcomes and patient safety (9).

The growing adoption of the ERAS protocol has heightened clinical attention toward the accurate assessment and effective reduction of POS. The assessment of shivering has gradually evolved from subjective scales to objective technologies. Among traditional evaluation tools, the methods established by Crossley and Mahajan primarily assess shivering intensity based on observations of peripheral vasoconstriction, peripheral cyanosis, and patterns of muscle activity. The Mathew scale emphasizes the evaluation of the intensity of fasciculations and muscle activity localized to the facial and cervical regions. Conversely, the bedside shivering assessment scale (BSAS) evaluates the distribution of shivering through visual inspection and palpation of the neck, chest, arms, and legs. The BSAS, with its established reliability and validity, has been widely adopted as an internationally accepted standard for evaluating shivering (10). Recent breakthroughs in sensor and artificial intelligence technologies have significantly advanced the objectivity and precision of shivering assessment. For instance, computer vision-based algorithms for analyzing muscle tremors can now detect characteristic rhythmic muscle contractions in real time, within a frequency range of 0.5–8 Hz. Studies have demonstrated that this approach can achieve an accuracy of up to 92.3% in identifying shivering episodes (11). Furthermore, by integrating multimodal data from infrared thermal imaging, accelerometers, and photoplethysmography (PPG), integrated wireless monitoring platforms enable comprehensive assessment and early warning of shivering. In the field of objective TCM research, Zhang Qian and colleagues investigated the use of deep learning to analyze TCM signs to assist in shivering recognition. They developed an intelligent tongue-pulse diagnosis system capable of identifying signs of Yang deficiency with cold congelation, such as a pale, swollen tongue and a deep, slow, and weak pulse. This system reportedly

demonstrates a predictive performance for shivering risk, with an AUC of 0.87 (12). This work paves a promising new avenue for integrating the principles of TCM pattern differentiation into the perioperative risk assessment framework (Table 1).

2.2 TCM perspective on postoperative shivering

Although “perioperative shivering” is not classified as a designated disease entity in TCM, its clinical presentation aligns with classical TCM disorders like “zhan li” (trembling) and “han jue” (cold limbs). The core pathology involves compromised Yang Qi, leading to inadequate bodily warming. The *Plain Questions: On Regulating the Meridians* elucidates the pathogenesis by stating, “Yang insufficiency results in external cold,” affirming that the deficiency of Yang Qi is the fundamental cause of cold intolerance and trembling. The “Treatise on Cold Damage Diseases” also repeatedly discusses the relationship between shivering and the stagnation or decline of Yang Qi. Surgical trauma and anesthetic agents are particularly prone to impairing the body’s vital Qi, especially Yang Qi. Zhang Jingyue of the Ming Dynasty stated in *Jingyue Quanshu*: “The Vital Gate (Mingmen) is the root of Primordial Qi (Yuanqi) and the abode of Water and Fire; the Yin Qi of the five Zang organs cannot be nourished without it, nor can the Yang Qi of the five Zang organs be activated.” This emphasizes that the fire of the Vital Gate is the foundation of the body’s entire Yang Qi. Postoperative depletion of Yang Qi and decline of the Mingmen fire result in an inability to warm the tendons and limbs, thereby leading to the onset of shivering.

Consequently, TCM identifies the etiology of postoperative shivering as fundamentally characterized by Yang Qi deficiency and weakened Mingmen fire, potentially compounded by excess manifestations such as cold congelation or blood stasis. The therapeutic principle is therefore to warm Yang, dispel cold, tonify the fire (of Mingmen), and support Yang. This approach is well exemplified by the classic formula Shen Fu Tang.

In addition to Yang Qi deficiency, constraint or stagnation of Yang Qi (Yang Qi Yu Zhi) represents another pivotal etiological pattern within TCM theory. The *Treatise on Cold Damage Diseases* delineates scenarios in which Yang Qi, although not deficient, becomes obstructed in its dispersing and warming functions, leading to impaired heat distribution and consequent shivering. This pattern often arises from pathogenic factors that disrupt the Shaoyang pivot mechanism or from internal obstructions by phlegm-dampness or blood stasis, which impede the free flow of Qi (13). In such cases, defensive Yang fails to properly reach the body surface, resulting in alternating chills and fever or shivering without true hypothermia. Within the perioperative context, factors such as anesthetic suppression, surgical trauma, emotional stress, or pre-existing Qi stagnation may contribute to a similar state of Yang constraint. The therapeutic strategy for this pattern shifts from pure supplementation to promoting Qi movement and unblocking stagnation, employing principles such as harmonizing Shaoyang and promoting Qi circulation, as embodied in formulas like Xiao Chaihu Tang (Minor Bupleurum decoction) (14). This broader conceptualization underscores that shivering in TCM is not monolithic but rather encompasses both deficiency and stagnation patterns of Yang dysfunction, thereby enriching the holistic diagnostic framework.

2.3 Advances in the understanding of shivering mechanisms in modern medicine

Shivering is a vital physiological response to hypothermia or pathological conditions and has traditionally been understood as being initiated by an elevation of the hypothalamic thermoregulatory set point. This elevation, in turn, triggers involuntary, rhythmic contractions of the skeletal muscles via motor pathways, a process aimed at generating heat and maintaining core body temperature. However, emerging evidence indicates that the mechanism of shivering is not confined to central regulation but rather involves a multilevel, networked physiological process encompassing peripheral tissue energy metabolism, mitochondrial function, and neuroendocrine regulation.

Mitochondrial complex I (CI), the key initiating enzyme in the electron transport chain responsible for catalyzing NADH oxidation, generates a proton gradient that serves as the fundamental driver of ATP production. During episodes of shivering, skeletal muscle energy demands increase dramatically, necessitating highly efficient ATP generation that is dependent on complex I (15). Research has demonstrated that dysfunction of complex I is a key contributor to insufficient heat production during shivering episodes, with the NDUF52 subunit playing a critical role. Located near the ubiquinone (CoQ) binding site, NDUF52 is essential for the assembly, stability, and catalytic activity of the complex. Downregulation or functional inhibition of NDUF52 severely impairs complex I activity, leading to reduced proton pumping efficiency, a decline in mitochondrial membrane potential, and ultimately compromised function of

TABLE 1 Comparison of POS assessment and early warning methods.

Method category	Specific method/ technique	Features
Traditional scoring scales	Crossley and Mahajan Scale	Simple and practical, yet susceptible to subjective bias
International Shivering Assessment Standards	Assessment of shivering distribution	Relatively accurate and detailed
AI-based predictive models	Artificial neural network (ANN)-based risk prediction model	Capable of processing multiple variables, with high predictive accuracy and strong clinical applicability
Real-time monitoring technologies	Computer vision-based muscle tremor analysis	Objective and real-time, with relatively high accuracy
Intelligent TCM assessment explorations	Intelligent tongue and pulse diagnosis system	Provides a novel approach for TCM pattern differentiation

ATP synthase (complex V), resulting in insufficient ATP supply (16). Under such conditions, even when a shivering response is present, skeletal muscle experiences an energy crisis that produces weak and ineffective contractions, manifesting as “shivering without heat production.”

Beyond the direct impairment of energy production, dysfunction of complex I can also induce severe secondary damage. When electron transfer is obstructed, electrons are prone to leak from specific sites within complex I (e.g., the I_Q site) and react with oxygen molecules, generating a surplus of superoxide and initiating oxidative stress. This not only exacerbates damage to mitochondrial structure and function, harming muscle cells, but also forms a vicious cycle with the ongoing energy crisis. Concurrently, the resulting imbalance in the NADH/NAD⁺ ratio disrupts multiple metabolic pathways, including glycolysis and the tricarboxylic acid (TCA) cycle, leading to global metabolic disturbances within the cell. Importantly, the body is not defenseless against a surge in ROS. In tissues such as skeletal muscle, uncoupling proteins (e.g., UCP3) can be activated to induce “mild uncoupling,” which lowers the mitochondrial membrane potential and thereby provides negatively feedback to suppress excessive ROS generation. This represents a crucial self-protective mechanism under cellular stress (17). Beyond the intracellular energy crisis, the systemic neuro-immune-inflammatory response also plays a critical role in the initiation and maintenance of POS.

This systemic metabolic disturbance further exacerbates shivering insufficiency through several direct and indirect pathways. First, impairment of glycolysis and the TCA cycle reduces the availability of pyruvate and NADH, which are key substrates for mitochondrial ATP synthesis. Consequently, even when shivering commands are centrally initiated, skeletal muscle faces an exacerbated energy crisis, leading to weaker and less effective thermogenic contractions (18). Second, the accumulation of metabolic intermediates—such as lactate and acyl-carnitines—alters intracellular pH and ion homeostasis, potentially disrupting excitation-contraction coupling in muscle fibers (19). Third, systemic metabolic dysregulation can influence the synthesis and release of neurotransmitters, including serotonin and catecholamines, which are integral to central thermoregulation and peripheral vasomotor tone (20). Thus, the metabolic consequences of complex I dysfunction not only undermine cellular energy production but also disrupt the integrated neuro-metabolic network required for effective shivering thermogenesis.

Recent studies have identified activation of the “brain-immune axis” as a core mechanism in the pathophysiology of POS, particularly for the inflammatory subtype. Specifically, surgical trauma and anesthetic agents can activate the “neuro-immune-inflammatory axis,” which disrupts the hypothalamic thermoregulatory set point through the following pathways.

The prostaglandin E2 (PGE2) pathway: This represents the primary pathway. Peripheral tissue injury activates cyclooxygenase-2 (COX-2), leading to substantial synthesis of PGE2 within the anterior hypothalamus. PGE2 binds to the EP3 receptors in the hypothalamus, which elevates the

thermoregulatory set point by inhibiting warm-sensitive neurons and exciting cold-sensitive neurons. This represents one of the most direct neuroimmune signals responsible for the trigger of shivering (21).

The HMGB1/TLR4/NF-κB pathway: High-mobility group box 1 (HMGB1), as a key damage-associated molecular pattern (DAMP), is released from injured cells and activates Toll-like receptor 4 (TLR4), thereby triggering the downstream nuclear factor kappa B (NF-κB) signaling. This cascade promotes the synthesis and release of pro-inflammatory cytokines, such as interleukin-6 (IL-6) and tumor necrosis factor-α (TNF-α) (22). These inflammatory cytokines not only exacerbate the systemic inflammatory response but can also act on the hypothalamus by crossing the blood-brain barrier or via vagal afferent signaling. As a result, the thermoregulatory set point is further elevated, generating an “inflammatory storm” that promotes shivering (23).

This neuro-immune-inflammatory axis interacts closely with the aforementioned mitochondrial-BAT thermogenic axis. On one hand, inflammatory signals such as TNF-α can directly inhibit the activity of mitochondrial complex I. Therefore, central immune-inflammatory signals orchestrate the initiation of shivering, while the functional state of peripheral mitochondria and brown adipose tissue (BAT) dictates the efficiency and effectiveness of thermogenesis. Together, these two components constitute the complete pathophysiological framework underlying POS.

In the body's defense against cold exposure, thermogenesis is achieved not only through shivering, which is primarily mediated by skeletal muscle contractions, but also through non-shivering thermogenesis mediated by uncoupling protein 1 (UCP1) in brown adipose tissue (BAT). Together, these mechanisms constitute a multilayered and complementary strategy for maintaining body temperature (24, 25). In contrast to ATP-dependent heat production in shivering, UCP1 operates by uncoupling substrate oxidation from ATP synthesis, dissipating energy directly as heat. Although the capacity for UCP1-dependent non-shivering thermogenesis in adults is relatively limited overall, it serves as an important supplement to shivering thermogenesis and may play a particularly significant role under specific conditions. A dysfunction in any component of either mechanism can lead to inadequate heat production under cold stress (26).

Furthermore, surgical trauma and other stress conditions can activate inflammatory responses, leading to the release of pro-inflammatory cytokines such as IL-6 and TNF-α. These cytokines act on the hypothalamus to alter the thermoregulatory set point, thereby promoting the initiation of shivering. In addition, neurotransmitter systems are involved. For instance, abnormal release of catecholamines and serotonin (5-HT) can affect central thermoregulation and peripheral vasoconstriction, thereby further modulating the intensity and duration of shivering (27).

In summary, shivering is a complex response initiated centrally and rely on the coordinated execution of peripheral energy metabolism and cellular/molecular functions. Greater

emphasis should be placed on multidimensional investigations into mitochondrial function and the neuro–endocrine–immune network to provide novel insights into the clinical prevention and management of shivering-related complications.

2.4 Dialog and integrative potential between traditional Chinese and Western medicine perspectives on shivering mechanisms

The “fire of the Mingmen” (vital gate) in TCM, considered the foundation of the body’s Yang Qi, serves a “warming function.” This physiological role aligns directly with the organism’s goal of maintaining core body temperature through heat production. Modern research has elucidated that mitochondrial energy metabolism—particularly the function of complex I—and UCP1-mediated non-shivering thermogenesis in brown adipose tissue represent central biological mechanisms for maintaining body temperature. Notably, Chinese herbal medicinals with functions such as “warming Yang to dispel cold and tonifying fire to reinforce Yang”—including Shenfu injection and cinnamon extract—have been shown to exhibit modern pharmacological actions that precisely target key nodes within these established modern thermoregulatory pathways. This convergence demonstrates that, despite employing distinct theoretical frameworks, TCM and Western medicine share a common understanding of the pathological essence of shivering and its therapeutic management. Such alignment provides a

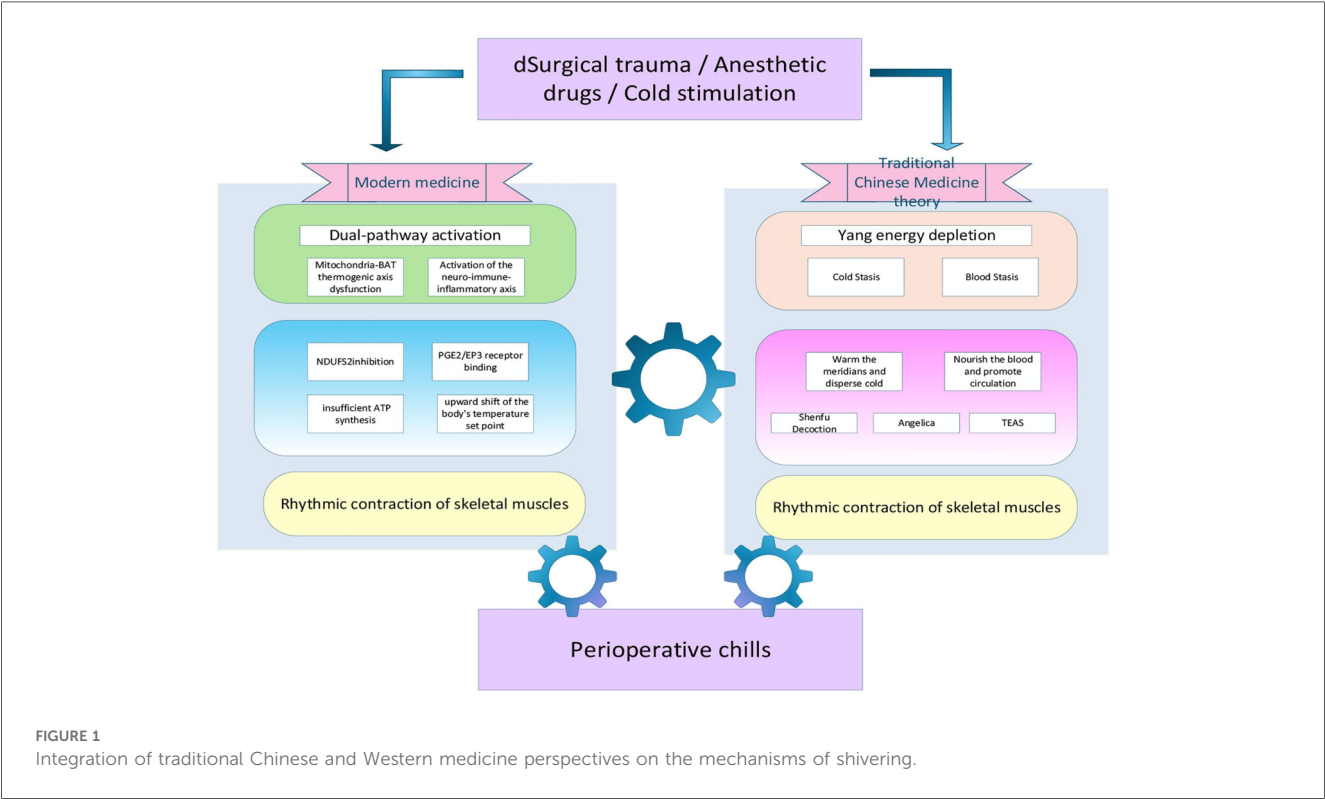
theoretical basis for their synergistic application, positing that Yang-warming herbs can simultaneously modulate both energy metabolism and neuroimmune responses, thereby enabling a multitarget regulatory strategy (Figure 1).

3 TCM-based prevention and management strategies for POS

TCM-based prevention and management of POS emphasizes “supporting Zhengqi and warming Yang, harmonizing nutritive and defensive Qi,” demonstrating advantages through multitarget mechanisms and holistic regulation. Available preventive and management strategies are diverse, encompassing internal herbal formulations, external techniques, and comprehensive care, thereby enabling flexible clinical selection tailored to specific syndrome patterns.

3.1 Internal herbal formulas and medicinals

The TCM approach emphasizes the core principles of warming and tonifying Yang Qi, dispelling cold, and unblocking the meridians. The classic formula *Shenfu Tang* (ginseng and aconite decoction), documented in the *Shi Yi De Xiao Fang* (*Effective Formulas from Generations of Physicians*), serves as a representative prescription. In this formulation, ginseng greatly tonifies primordial qi, while aconite warms the kidney and reinforces Yang, thereby jointly restoring Yang and rescuing



collapse. Modern research has not only validated its classical efficacy but also further elucidated its scientific basis at the core level of energy metabolism. Studies have shown that the active components of Shenfu injection upregulate the expression of sirtuin 1 (SIRT1), thereby activating the downstream key factor peroxisome proliferator-activated receptor gamma coactivator-1 α (PGC-1 α). This signaling cascade effectively promotes mitochondrial biogenesis and enhances the functional activity of mitochondrial respiratory chain complexes (28). As a master regulator of mitochondrial biogenesis, PGC-1 α activation significantly promotes the generation of new mitochondria and enhances the functional activity of mitochondrial complex I, thereby efficiently elevating cellular ATP production (29). This mechanism, which targets the cell's power plant (mitochondria), provides solid molecular evidence for the TCM theory of “tonifying the fire of the Mingmen (vital gate).” The SIRT1/PGC-1 α signaling axis effectively bridges the ancient concept of reinforcing primordial Yang with contemporary mitochondrial biology, demonstrating how Yang-warming herbs enhance energy metabolism at the subcellular level.

For patterns characterized by cold congelation and blood stasis, *Danggui Sini Tang* (*Angelicae sinensis* decoction for frigid extremities), with appropriate modifications, can be selected to warm the meridians, dispel cold, nourish the blood, and unblock the vessels. Modern pharmacological studies reveal that the therapeutic effects of this formula extend beyond simple warming and unblocking effects, directly targeting the neuro-immune-inflammatory core implicated in the pathogenesis of POS. Research has demonstrated that *Danggui Sini Tang* not only significantly reduces plasma prostaglandin E2 (PGE2) and serum interleukin-6 (IL-6) levels—effectively inhibiting the inflammatory upward shift of the central thermoregulatory set point—but also ameliorates microcirculatory dysfunction, thereby ensuring adequate energy supply to peripheral tissues and facilitating the clearance of metabolic waste products (30). Through these two synergistic dimensions—anti-inflammatory effects and improved tissue perfusion—the modern scientific basis underlying the formula's classical function of “warming the meridians and dispelling cold” is collectively elucidated.

3.2 External therapies in TCM

As a distinctive external therapy in TCM, TEAS has recently become a research focus due to its non-invasive nature, proven efficacy, and ability to alleviate perioperative complications, including anxiety, gastrointestinal dysfunction, and pain (31). A frequently selected acupoint is Zusanli (ST36). As the He-Sea point of the stomach meridian, it can be stimulated by both moxibustion and acupuncture to warm the middle jiao, tonify deficiency, and harmonize the Qi and blood (32). This demonstrates that, despite employing distinct theoretical frameworks, TCM and Western medicine share a common understanding of the pathological essence of shivering and its therapeutic management. Such alignment provides a theoretical

basis for their synergistic application, positing that Yang-warming herbs can simultaneously modulate both energy metabolism and neuroimmune responses, thereby enabling a multi-target regulatory strategy: At the supraspinal level, signals generated by TEAS at ST36 are transmitted via A β and A δ primary afferent fibers, which subsequently ascend to activate the key descending inhibitory pathway comprising the periaqueductal gray (PAG) and the rostral ventromedial medulla (RVM). This activation leads to the release of serotonin (5-HT) and norepinephrine to the spinal dorsal horn, which effectively suppresses the ascending transmission of noxious thermal signals, including those responsible for shivering (33). At the systemic immunomodulatory level, TEAS has been established as a potent activator of the “cholinergic anti-inflammatory pathway.” Afferent signals are transmitted via the vagus nerve, promoting the release of acetylcholine from nerve terminals. The subsequent binding of acetylcholine to the $\alpha 7$ nicotinic acetylcholine receptor ($\alpha 7$ nAChR) on immune cells, such as macrophages, inhibits activation of the nuclear factor kappa-B (NF- κ B), thereby significantly suppressing the release of pro-inflammatory cytokines, including tumor necrosis factor- α (TNF- α) (34). This mechanism provides a key explanation for the systemic and long-lasting anti-inflammatory and anti-shivering effects of TEAS.

3.3 Personalized strategies based on syndrome differentiation

Traditional Chinese medicine emphasizes the principle of “tailoring treatment to the individual, season, and locality” (three factors considerations). In clinical practice, personalized interventions must be implemented based on the patient's constitution and syndrome type. For patients with Yang Qi deficiency syndrome, characterized by an aversion to cold, cold limbs, pale complexion, and a deep-slow pulse, a recommended intervention is intravenous infusion of Shenfu injection combined with ginger-partition moxibustion at the Shenque (CV8) acupoint (35). For patients presenting with cold coagulation and blood stasis, which is characterized by cyanotic lips, cold extremities, and a purplish-dark tongue, a suitable therapeutic approach involves the use of modified *Danggui Sini* decoction complemented by pricking and cupping at the Xuehai (SP10) acupoint (36); Postoperative shivering in patients after cesarean section is primarily attributed to impairment of the Thoroughfare and Conception Vessels, characterized by Yang deficiency and blood depletion. The recommended treatment principle is to warm the channels and nourish the blood, employing moxibustion at Sanyinjiao (SP6) and Qihai (CV6) acupoints, combined with the oral administration of herbal formulas that warm the meridians and replenish blood (37). Through the synergistic integration of pharmacological and non-pharmacological approaches, TCM simultaneously addresses systemic regulation and local intervention, thereby demonstrating distinctive strengths in the prevention and treatment of POS.

4 Perioperative shivering: contemporary approaches in Western medicine

The Western approach to preventing and managing POS primarily employs a dual-approach strategy of pharmacological intervention and temperature management. In recent years, the field has been evolving toward precision, personalization, and combined multimodal strategies.

4.1 Pharmacological strategies for prevention and management

The pharmacological strategy for managing POS in Western medicine has evolved from single-symptom control to multitargeted, precision-based interventions that address its complex mechanisms. In this context, the application of dexmedetomidine epitomizes the current cutting edge.

4.1.1 Dexmedetomidine: A multi-mechanism agent at the forefront

As a highly selective α_2 -adrenoceptor agonist, dexmedetomidine has emerged as a first-line medication due to its potent anti-shivering efficacy and favorable safety profile. Its therapeutic actions extend beyond sedation to encompass a core advantage: significant anti-inflammatory properties. Studies demonstrate that dexmedetomidine effectively inhibits the release of damage-associated molecular patterns (DAMPs), such as HMGB1. This action thereby blocks its binding to Toll-like receptor 4 (TLR4) and downregulates the activity of the downstream NF- κ B signaling pathway, ultimately leading to a significant reduction in the production of key pro-inflammatory cytokines, including IL-6 and TNF- α (38). Based on this mechanism, its clinical application strategy has shifted from “post-occurrence management” to “preventive intervention.” Evidence-based medicine indicates that initiating a preventive continuous infusion at a low dose (e.g., 0.2–0.4 μ g/kg/h) during surgery or in the early post-anesthesia recovery phase can more effectively cover the peak period of inflammatory factor release, thereby achieving superior preventive efficacy while avoiding the hemodynamic fluctuations associated with a single bolus dose (39, 40). This regimen is particularly suitable for elderly patients or those with cardiac insufficiency, whose circulatory systems have limited compensatory capacity (41).

4.1.2 Prophylactic medication: Targeting the Cause

Preoperative administration of glucocorticoids, such as dexamethasone, exerts potent anti-inflammatory effects by inhibiting the COX-2/PGE2 pathway. This mechanism reduces the inflammatory-induced upward shift of the central body temperature set point and has been confirmed by multiple studies to significantly reduce the incidence of shivering following medium-to-large surgeries (42). Similarly, the prophylactic use of non-steroidal anti-inflammatory drugs (NSAIDs), such as parecoxib sodium or flurbiprofen axetil, also

acts by inhibiting prostaglandin synthesis. This shared mechanism establishes their role as integral components of multi-modal strategies for both analgesia and shivering prevention (43, 44).

Remimazolam, a new benzodiazepine, holds considerable promise for refined anesthesia regimens. Its profile is distinguished by limited inhibition of the mitochondrial respiratory chain and, critically, a unique esterase-based metabolic pathway that underpins its rapid pharmacokinetic profile and superior hemodynamic stability. These features enable its synergistic use with opioids, reducing opioid dosage and consequently the shivering risk, which translates into an enhanced sedation strategy for vulnerable populations (45).

4.1.3 Adjunctive and traditional pharmacotherapy: promise and challenges

Although dexmedetomidine has become mainstream, other drugs remain relevant in specific contexts. However, their use involves inherent tradeoffs, underscoring the inadequacy of single-pathway interventions.

Opioids such as fentanyl and butorphanol can rapidly suppress shivering by acting as kappa-opioid receptor agonists; however, their associated risks of respiratory depression and sedative effects are often contraindicated within the ERAS protocol, relegating their use primarily to rescue therapy after the onset of shivering (46). Tramadol was previously favored due to its dual mechanism involving inhibition of serotonin (5-HT) and norepinephrine reuptake, but its clinical utility is limited by a high incidence of nausea and vomiting, which significantly compromises patient comfort (47). Magnesium sulfate, acting as an NMDA receptor antagonist, demonstrates favorable cost-effectiveness and safety in specific populations, such as patients undergoing cardiac surgery; however, the consistency of its efficacy requires further validation (48).

A contemporary re-examination of the dopaminergic system has clarified that the principal anti-shivering mechanism of the traditional agent pethidine lies in its action as a central dopamine D2 receptor agonist. However, activation of receptors within the widely distributed chemoreceptor trigger zone means that shivering suppression comes at the cost of a significantly elevated risk of nausea and vomiting, which considerably restricts its clinical application (49). The role of the hypothalamic orexin system in the integrated control of the intricately linked processes of wakefulness and thermoregulation has garnered significant attention (50). Pervasive inhibition of the orexin system by general anesthetics is thought to contribute to delayed restoration of the central temperature set point during emergence, thus identifying it as a compelling target for future pharmacological interventions (51).

It follows that these traditional pharmacological options are limited by their focus on either immediate symptom suppression or isolated pathways, limiting their ability to deliver a holistic and fundamental therapeutic effect that is often required. This limitation, in turn, highlights the necessity and distinct advantage of establishing a combined TCM–Western medicine paradigm, centered on dexmedetomidine and

synergistically incorporating the holistic principles of TCM (Figure 2).

4.1.4 Active temperature control and insulation

The core concept in current perioperative temperature management has undergone a paradigm shift, moving away from reactive, compensatory warming toward a proactive defense system centered on “preventive warming” and “precision monitoring.” This strategic shift is driven by growing recognition of the predictable decline in core body temperature that occurs following the induction of anesthesia.

Preoperative warming is one of the most effective preventive warming strategies. It is recommended to actively warm the patient’s body surface using a forced-air warming blanket for at least 30 min before surgery, with the aim of raising core body temperature to above 36.5 °C (7). This preemptive measure establishes an adequate body heat reserve before anesthesia-induced vasodilation and internal heat redistribution, thereby significantly attenuating both the magnitude and rate of subsequent temperature decline (52).

Precise temperature management inherently depends on accurate monitoring techniques. The maturation and widespread adoption of non-invasive, continuous core temperature monitoring technology over the past few years have made this goal attainable (53). Technologies such as zero-heat-flux and dual-sensor methods overcome the inherent lag and inaccuracy of traditional axillary temperature monitoring, enabling real-time, continuous tracking of core temperature during surgery. This allows anesthesiologists to dynamically evaluate warming efficacy and adjust strategies promptly, truly achieving a shift from “empirical” to “data-driven” precision temperature management. Guided by this principle, the integrated application of established measures—such as forced-air warming blankets, fluid warmers, and ambient temperature

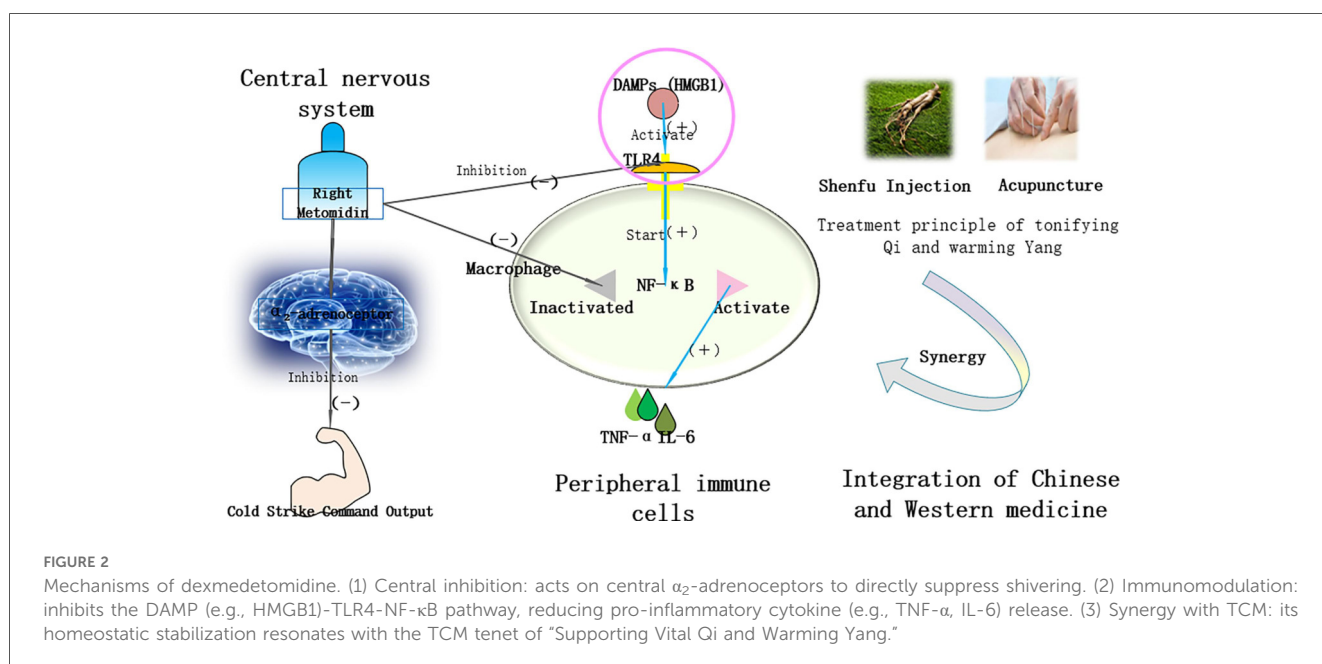
control—during surgery is essential to achieve the greatest synergistic benefit (54).

5 Integrating TCM and Western medicine: clinical strategies and therapeutic approaches

The management of POS has evolved from single-modality interventions toward integrated, multi-disciplinary, and multimodal strategies. Building on the previously outlined mechanistic understanding of shivering from both traditional Chinese and Western medicine, we propose an integrated clinical model centered on the core principles of “disease–syndrome combination, dynamic–static interaction, and prophylaxis–therapy integration.” This model emphasizes achieving complementary advantages between both medical systems through precise assessment, ultimately improving patient outcomes.

5.1 Trinity integrated protocol: assessment, early warning, and intervention

Starting with the integration of modern monitoring technologies and TCM diagnostic methods, this system achieves dynamic perception and early warning of shivering risk. By combining an artificial intelligence-based myotremor analysis system that detects rhythmic muscle contractions (0.5–8 Hz) with an intelligent tongue-pulse diagnostic device that identifies TCM signs of Yang deficiency (e.g., pale, swollen tongue, and a deep, slow, weak pulse), the model significantly enhances the sensitivity of shivering risk prediction. Once a high risk is flagged, the early warning system initiates multidimensional interventions.



A proposed synergistic drug strategy recommends a regimen combining dexmedetomidine with Shenfu injection. Prophylactic administration may include a low dose of dexmedetomidine (e.g., a small loading dose followed by continuous infusion at a low maintenance rate) during surgery or in the postanesthesia care unit, alongside intravenous infusion of Shenfu injection at standard clinical dosages. The specific dosing regimen and timing should be individualized based on the patient's body weight, physiological status, and intraoperative conditions (55). Dexmedetomidine exerts its antishivering effect by activating central α_2 -adrenoceptors, whereas Shenfu injection, leveraging its components including ginsenosides, acts via multiple pathways. On one hand, it upregulates UCP1 expression via the β_3 -AR/cAMP pathway to enhance non-shivering thermogenesis. On the other hand, its holistic “warming Yang” effect contributes to the improvement of cellular energy metabolism. Neuro-immune modulation: Research has shown that applying 2/100 Hz TEAS at the Zusanli (ST36) acupoint (56) can activate the hypothalamic arcuate nucleus–nucleus tractus solitarius pathway, promote the release of endorphins, and inhibit the TLR4/NF- κ B signaling pathway in the spinal dorsal horn. This thereby effectively reduces the levels of pro-inflammatory cytokines IL-6 and TNF- α , thus blocking the onset of inflammation-mediated shivering (57, 58).

Enhanced physical warming: A combination of a constant-temperature garment and an Orve+ wrap with a warming blanket is used, beginning with 30 min of prewarming. This approach establishes a dual thermoregulatory mechanism of “passive heat storage plus active regulation,” significantly reducing the incidence of hypothermia and associated shivering (59).

5.2 Emerging pathway for combating mitochondrial dysfunction

Addressing the key pathology of anesthetic-induced mitochondrial complex I dysfunction, we propose an integrated TCM–Western medicine intervention strategy centered on “preserving NDUF2 subunit function and promoting non-shivering thermogenesis.” Studies have shown that aconite alkaloids and ginsenosides in Shenfu injection can partially reverse the suppression of mitochondrial function induced by anesthetics such as sevoflurane, thereby consolidating the “vital gate fire” at the molecular level (60). Concurrently, the novel benzodiazepine remimazolam, which exerts minimal inhibition of the mitochondrial respiratory chain and is metabolized independently of the P450 enzyme system, can be combined with Shenfu injection. This combination reduces the total sedative dose required, thereby lowering the risks of hypotension and respiratory depression, making it particularly suitable for elderly and frail patients (45). Moreover, cinnamon extract (cinnamaldehyde) activates the TRPV1 channel, directly stimulating brown adipose tissue and upregulating UCP1 expression by approximately 2.1-fold. This mechanism provides an alternative thermogenic pathway that bypasses mitochondrial complex I (61), offering a pharmacological basis for the modern

application of the TCM principle “warming Yang to disperse cold.”

6 Conclusion and prospects

Perioperative shivering is a common complication mediated by complex networked mechanisms, and its pathophysiology extends far beyond a simple thermoregulatory imbalance. Modern research elucidates that the activation of the neuro-immune-inflammatory axis and dysfunction of peripheral mitochondrial bioenergetics are its pivotal mechanisms. In parallel, the TCM theoretical framework of “Yang Qi deficiency and vital gate fire decline” offers a unique holistic perspective for understanding this phenomenon. In terms of prevention and management, Western medicine relies on targeted pharmacological agents like dexmedetomidine and active temperature management strategies, while TCM, exemplified by Shenfu injection and TEAS, exerts its unique advantages through multitarget actions and holistic regulation. The integrated “assessment–early warning–intervention” model that combines TCM and Western medicine represents a frontier approach in this field. By integrating AI-powered objective assessment with TCM-based intelligent syndrome differentiation, it enables early identification of shivering risk. Through the synergy of interventional measures from both medical systems, this approach aims to achieve precise modulation across multiple levels, including central nervous regulation, immune homeostasis, and energy metabolism. Looking forward, further exploration is needed to elucidate the molecular and cellular mechanisms of shivering from both TCM and Western medicine perspectives. Rigorous clinical trials are required to validate the efficacy and safety of this integrated approach, followed by the establishment of standardized operating procedures to promote its widespread application in perioperative management.

Author contributions

CL: Data curation, Formal analysis, Investigation, Methodology, Project administration, Validation, Visualization, Writing – original draft, Writing – review & editing. MZ: Data curation, Investigation, Methodology, Validation, Writing – original draft. R-nC: Data curation, Investigation, Methodology, Validation, Writing – original draft. YW: Data curation, Investigation, Methodology, Validation, Writing – original draft. ZL: Conceptualization, Funding acquisition, Project administration, Resources, Supervision, Visualization, Writing – review & editing.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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The author(s) declared that generative AI was not used in the creation of this manuscript.

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